

Sudhakara reddy G

Proven Expertise in Automotive Seating & Restraints Engineering Manager | Automotive Safety Systems

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PROFESSIONAL SUMMARY

Results-driven Seating and Restraints Engineering Manager with 9 years of experience leading full-system development from concept to SOP. Expertise in managing cross-functional teams to deliver complete seat systems (frames, foam, trim, mechanisms) and occupant safety integration. Proven track record of resolving complex integration conflicts between comfort, weight, cost, and regulatory compliance (FMVSS/ECE). Adept at driving DFSS initiatives and managing Tier 1 supplier relationships to achieve zero-defect launches.

CORE COMPETENCIES

- **Technical Design and Engineering** DFMEA, DVP&R, GD&T, CAE Correlation, Homologation CAE Correlation, Homologation.
- **Seating-Specific Expertise:** Ergonomics (ISO 16121), Occupant Safety (ECE R14, R80, R118, FMVSS 207/210/302), Foam Formulation & Comfort Tuning, Mechanism Design (Recline, Sliding, Lift), Anti-submarining, Fire Retardancy (ECE R118, FMVSS 302).
- **Product Development:** APQP/PPAP, DFM/DFA, Value Engineering/VA-VE, Design Validation (DV), Production Validation (PV), Root Cause Analysis (8D, 5Why), OEM Requirements Management, Complete Seat Assembly, Seat Structures, Head Restraints, Airbag Integration, Seatbelts, Interior Trim
- **Regulations:** FMVSS 201/202/207/210, IIHS Whiplash, ECE R17/R14, NCAP
- **Engineering:** DFMEA, DVP&R, GD&T, CAE Correlation, Homologation
- **Management:** Budgeting, CAPEX, Resource Planning, Global Platform Launches
- **Project Management:** Cross-functional Leadership, Supplier Technical Management, Cost Engineering, Timeline Management.
- **Systems:** Complete Seat Assembly, Seat Structures, Head Restraints, Airbag Integration, Seatbelts, Interior Trim

PROFESSIONAL EXPERIENCE

Assistant Manager at Meen Motors-India – (Renault Alpine DR110 May 2025– Present)

Lead a team of 8 senior engineers and designers responsible for the design, validation and launch of seating systems for Renault Alpine Sport Ev Roofless 2 row Buckets Seats and With Convertible Roof 4-seater

- Led the structural design of a new high-end touring coach seat, utilizing FEA to optimize a composite backshell structure, resulting in a 22% weight reduction versus the previous model while maintaining ECE R14 strength requirements
- **Program Leadership:** Directed the successful launch of DR110, managing the program budget. Delivered seats under mass target while achieving a 5 rating in IIHS head restraint evaluations.
- **Systems Integration:** Resolved critical headroom clearance and airbag deployment trajectory conflicts between Seating and Roof/Pillar trim teams, preventing a 3-month program delay.
- **Validation & Compliance:** Owned the DVP&R for all occupant protection systems. Achieved 100% first-pass pass rate on FMVSS 210 (Seat Belt Anchorages) through targeted CAE correlation and component-level testing.
- **Supplier Management:** Led strategic sourcing initiative for recliner mechanisms, negotiating a 4000\$ cost reduction and improving durability cycle testing by 100%.
- **Mentorship:** Implemented a formal DFMEA training program for the department, reducing late-stage change requests by 30%.
- **Concept to SOP:** Engineered the front seat frames for DR110, focusing on high-strength steel applications to meet crash pulse requirements without increasing wall thickness
- **Problem Solving:** Led root cause analysis on seatbelt buckle rattling and seat creak noise issues. Implemented a production line shim adjustment process that reduced warranty claims by 20% within 6 months
- **Cross-Carline:** Managed carry-over part strategy across 3 different vehicle models, resulting in engineering cost avoidance.
- **CAE Collaboration:** Worked closely with CAE leads to optimize shoulder belt mounting points, achieving a 15% improvement in torso restraint kinematics.
- Customized designs and protocols to satisfy customer needs and regulatory standards
- Led team meetings, fostering effective communication amongst team, Tier-1 Supplier, OEM, Serial Production, SOP, JIT.
- Streamlined workflow by implementing efficient task management strategies by using JIRA and click-up.
- BOM Cost, Tooling Investment, Piece Price considering while Releasing the Ecr
- Prototyping, Soft/Hard Tooling, Pilot Run
- Collaborated with cross-functional teams, integrating engineering expertise with other areas of knowledge to achieve common goals.

Assistant Manager at Renault Nissan and Business Center (Sept 2024 – May 2025)

- Lead the design and development of the Duster DDE Centre Arm Rest and Seat Structures, including benchmarking, packaging studies, and ECR (Engineering Change Request) management.
- Collaborated with cross-functional teams, including interior/exterior styling, project management, and production teams, to ensure on-time delivery of design releases.
- Utilised CATIA, 3D Experience to design and package occupant systems, ensuring compliance with FMVSS, ECE, and ARAI regulations.
- Developed seating layout assembly drawings, homologation documentation, and labelling standards compliant with various international regulations.
- Managed engineering bill of materials (EBOM) ensuring accuracy and completeness throughout product lifecycle.
- Managed product development from concept through to program launch, ensuring design feasibility and addressing client specifications.
- Drove innovation and contributed to the filing of several team patents, enhancing product performance and differentiation.

Lead Project Assistant Manager – Passive Safety Systems at ZF Lifetec (Aug 2022 – Sept 2024)

- Led design and development of seat belt assemblies, airbag housing, and steering wheel, ensuring compliance with global safety standards for electric vehicles.
- Delivered seat belt retractor mounting fixture designs, steering wheel, and airbag packaging in alignment with customer requirements (BMW, Mercedes, JLR).
- Provided manufacturing plant support and led change management activities, including resolving workflow issues through Jira.
- Facilitated Teamcenter and Windchill releasing activities for timely project completions.
- Worked extensively on APQP and PPAP documentation to support successful product launches.
- Collaborated with OEMs and suppliers to execute product validation and support certification processes across global markets (Europe, China, and the US).
- Seat belt principal video creation using 3ds max Animation creation for Retractor sensor locking, belt and Spool movement at the time vehicle collision and Material selection for rendering better visualization to showcase the seat belt technology to customers.
- Steering wheel trims, Switch bezel, Back cover 3d modelling and fixation clip with respect to Steering column, 2d drawings creation
- Airbag housing plastic design and development, Considering the cae team result.

Seat Sr. Design Engineer – Mechanical at Tata Technologies (Aug 2021 – Aug 2022)

- Led the CAD conversion and design of automotive Commercial seating arrangements and 3d modelling, 2d creation of Seating layout for OEM clients JLR L492 and L482
- Focused on surfacing and packaging design, ensuring components met vehicle interior specifications and manufacturing constraints.
- Delivered parts and assemblies on time, supporting the creation of production-ready designs and validation.
- Specialized in design and development of plastic interior trims including seating Foam, Seat Shell, Armrest, Toolbox and detailed annotations.
- Seating arrangement on Chassis, as per regulation following by ECE R14, R17, 21 and 3d Modelling and Drafting as per Tata requirement Product life cycle release activity.
- Managed exterior trim designs including door cladding, front and rear fender components.

Seat Research Engineer at Hyundai Transys (Feb 2019 – Aug 2021)

- Involved in seating layout design, homologation processes, and updating label drawings to meet country-specific standards.
- Designed critical seating system components including height adjuster levers, recliner levers, and backboard designs for automotive seating.
- Engaged in DFMEA (Design Failure Mode Effects Analysis) with senior engineers to assess risk and improve seat system designs.
- Supported H-point testing and contributed to ARAI crash analysis, ensuring compliance with local safety standards.
- Side valance cover, Inner plastic cover, calf support plastic trim, Mtg plastic trims with respect to carpet and Sheet metal mtg brackets.
- Good understanding on manufacturing feasibility & Assembly processes and Tooling process automotive seats. Familiar knowledge of ARAI and ECE regulations.
- Familiar knowledge Foam Testing's and H-Point Measurement in vehicle level, Foam Manufacturing of seating system.
- Business Trip plant visit on Resolving of seat Plant Design and development, Quality issues from Customer side

Jr. Design Engineer/Mechanical Engineer at Hinduja Tech (Client- Ashok Leyland Commercial E Bus Pvt Ltd-June 2018 – Feb 2019)

- Designed front and rear windshields, side windows, and guard rails for bus body applications, using CATIA for 3D modelling and detailing.
- Managed product development from concept to final detailing, ensuring all components met design specifications and manufacturing constraints.
- Worked closely with the assembly team to ensure proper fitting and bonding of window and Front fascia components.
- Developed 3D models and detailed assembly drawings, including annotations and parting line designs for plastic components.

Trainee Engineer at Ascent Technologies (Client- Larsen Toubro Pvt Ltd-April 2017 – June 2018)

- Hydraulic Piping design for ship building layout drawing
- Gd & T training, Manufacturing Sheet metal, Casting, Plastic design and Development learning
- Reverse Engineering of the component.
- Machining components quality check using measurements tools

EDUCATION

- **Bachelor's Degree in Mechanical Engineering** | Lakireddy Bali Reddy College of Engineering, India
- **Diploma in Mechanical Engineering** | Bapatla Polytechnic College, India

SKILLS

- Tier 1 sourcing
- Program Lifecycle (Gateways)
- Executive Presentations
- Conflict Resolution (between attributes like Cost vs. Comfort)

ADDITIONAL SKILLS

- Occupant Safety / Crashworthiness
- Whiplash mitigation
- H-point control
- Trim & Foam Development
- Mechanism Integration (Recliners, Sliders, Risers, Latches)
- Wire Harness routing (for heated/cooled seats)
- Airbag Chute design
- EBOM Management, Experienced in Engineering Bill of Materials management ensuring accurate product data management.